

Basic Immunology

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What is Immunity?

- Protection from disease and especially infectious disease
- Immune system
 - Cells and molecules involved in such protection
- Immune response
 - response to introduction of a foreign agent

Autoimmune diseases

- individuals mount immune responses to their **own tissues** as if they were foreign agents

Cells of the immune system

- Pluripotent stem cell (bone marrow)
 - lymphoid progenitor
 - myeloid progenitor
- Lymphoid progenitor → B lymphocyte
 - T lymphocyte
 - NK cell

- Myeloid progenitor→
 - immature dendrite cell→ mature
 - granulocytic/ macrophages progenitor→
granulocytes/ macrophages
 - megakaryocytic/ erythrocyte
progenitor→ platelet/ RBC

Organs of the immune system

- Primary (**central**) lymphoid organs
- Secondary (**peripheral**) lymphoid organs

Central lymphoid organs

- Bone marrow and thymus
- Bone marrow → formation of pluripotent stem cell
→ B lymph. Maturation
- Thymus → T lymph. Maturation before contact with Ag.

Peripheral lymphoid organs

- Lymph nodes
- Spleen
- Mucosal associated lymphoid tissue (tonsils, peyer's patches, appendix etc.)
- Sites where **Ags are trapped** and lymphocytes encounter Ag and differentiate into **effector cells**

- L. node → collect Ag from site of infection in tissue
- Spleen → collect Ag from blood
- MALT → collect Ag from epithelial surfaces

General features

- Two broad categories
 - 1) Innate immunity (non specific)
 - 2) Adaptive immunity (specific)

Innate immunity

- 1st line defense
- Immediate action
- Born with us

Innate immunity

- No prior exposure (priming) needed
- Acts in *same way* in all normal individuals
- *A defining aspect of the innate immune system is that it carries no memory of an encounter with a foreign organism.*

- Made up of several components.
- **Physical barriers** → skin and mucous membranes
- **Physiological factors** → pH, temperature and oxygen tension
- **Protein secretions** → lysozyme, complement, interferons, C-reactive protein

Cells

- **Phagocytic cells** are critical in the defense against bacterial and simple eukaryotic pathogens
- Includes **neutrophils** and **macrophages**.

- Polymorphs
- Macrophages
- Basophils
- Mast cells
- Natural killer cells (NK cells)

- Macrophages and polymorphs- phagocytic cells
- Control common infections but cannot always eliminate infectious agents
- Initiate and direct adaptive immune response

- E.g. bacteria → phagocytes engulf and secrete cytokines → inflammatory response i.e. effect on blood vessels (calor, dolor, rubor, tumor)

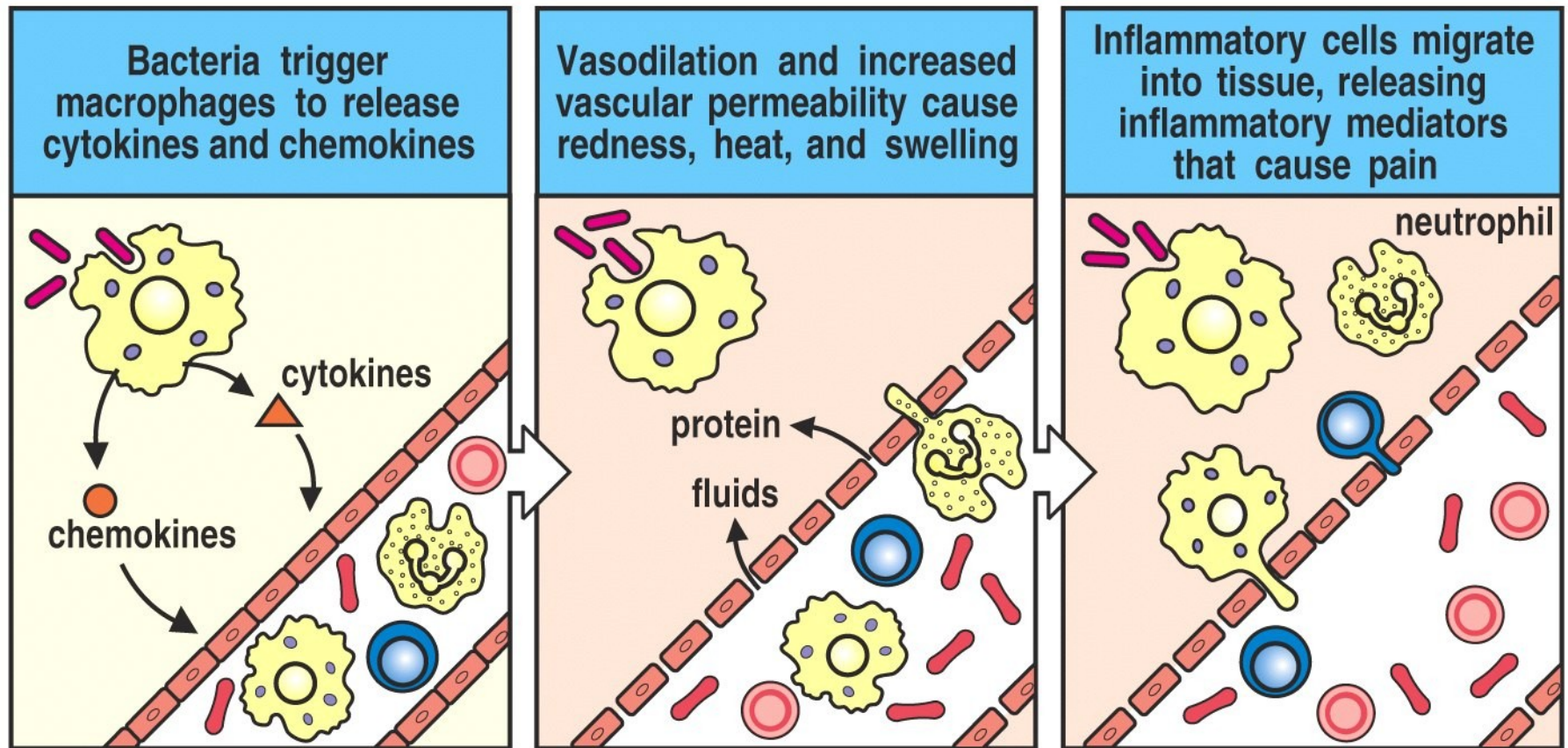


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Adaptive immunity

- 2nd line of defense
- 4-5 days after infection
- Occurs during lifetime
- Response specific to the organism
- Memory → rapid recall system
 - stronger reaction to re infection
- Mediated by lymphocytes

Types of Adaptive immunity

1. Humoral immunity → used for pathogens outside cells e.g. bacteria
2. Cellular immunity → used for pathogens inside cell e.g. virus, intracellular bacteria

Humoral immunity

- Mediated by B lymphocytes
- B cell mature → plasma cells → antibody
- Antibodies are proteins → immunoglobulin
- Antibody combines with antigen
- Ag-Ab complex → lock and key reaction between molecules of complementary configuration

Antibody

- bind antigen
- generally highly specific
- are only produced by B lymphocytes

Antibody

- found in the plasma, secretory fluids such as mucus, milk and sweat
- All antibodies have a similar overall structure with two light and two heavy chains

Actions of Abs

- **Neutralization**
 - bind to pathogens
 - cover harmful areas
 - block access of bacteria to cells
- **Opsonization** → enables phagocytosis
- **Complement activation** → Ab + bacteria → lysis by complement

Classes of antibodies (isotypes)

- 5 types
- IgM, IgG, IgA, IgE, IgD
- IgM → 1st Ab produced
 - action → complement activation

Classes of antibodies (isotypes)

- **IgG** → most abundant type in blood
-action → neutralization, opsonization, complement activation
- **IgA** → body secretions, mucus epithelium (intestinal, respiratory)
-action → mainly neutralization

- **IgE** → bound by receptors on mast cells
beneath skin, mucosa and along blood vessels
in connective tissue

-actions → **trigger mast cells** → inflammatory
mediators

induce coughing, sneezing, vomiting

- IgD → expressed on surface of B cells
-action → not clear

Cell mediated immunity

- Intracellular organism not accessible to Ab
- Dependent on T lymphocytes
- Ag+ T cells → clonal proliferation
 - effector T cells
 - memory T cells

Effector T cells

- 2 main groups

1. Cytotoxic T cells → CD8 cells

2. Helper/ inflammatory T cells (Th1,Th2)
CD4 cells

Cytotoxic T cells

- Usually CD8 cells
- **Directly lyse target** cells e.g. virus infected cells
- Important against **viral infections**

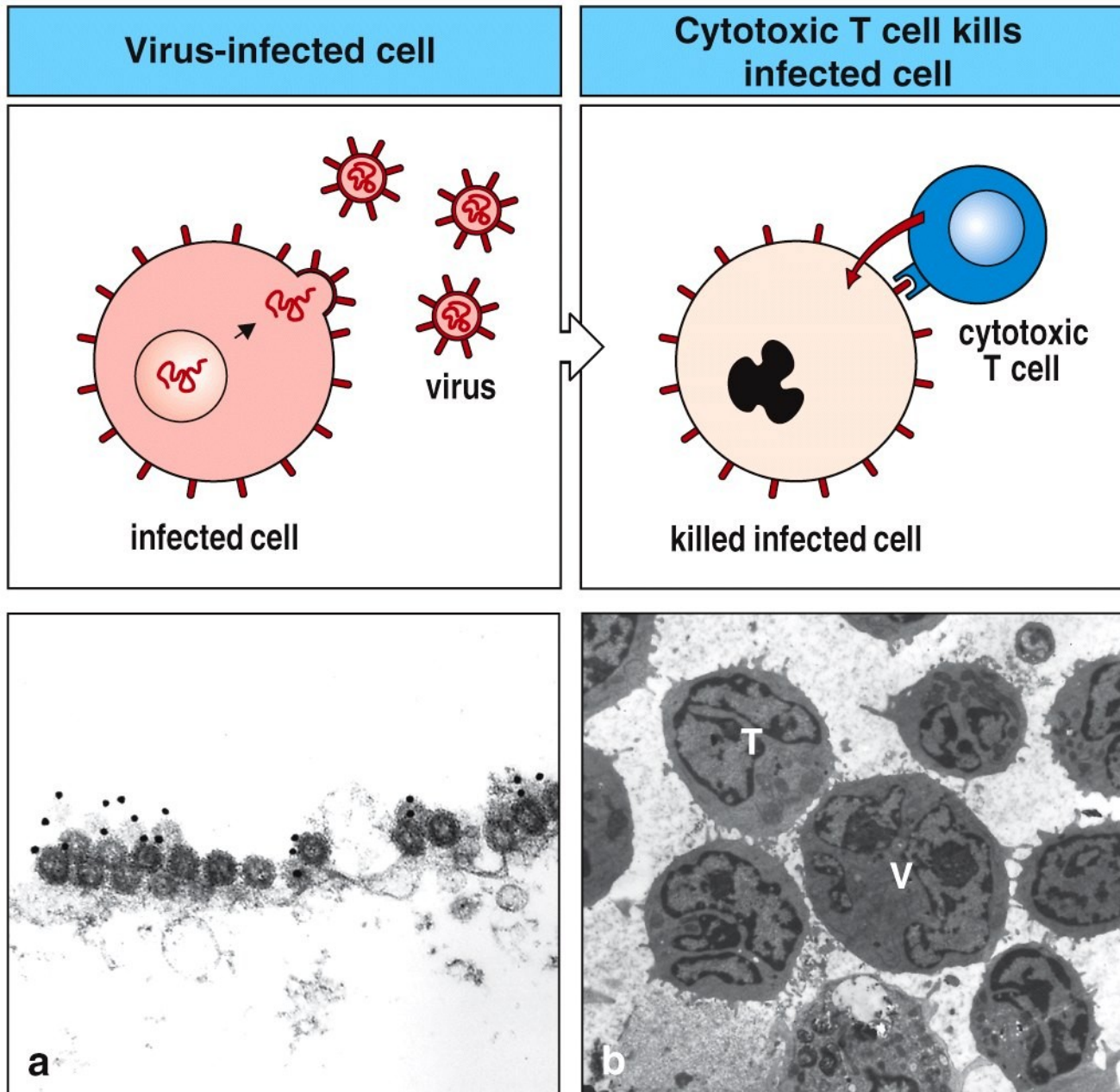


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Helper/ inflammatory T cells

- Usually CD4 T cells
- Specialized to activate other cells
(macrophages, B lymphocytes)

- Two functional classes
 1. TH1 cell (inflammatory T cell)
activate macrophage to kill intracellular organism
 2. TH2 cell (helper T cell)
activate B cell to make Ab

Hypersensitivity reaction

- **Fine balance** b/n host under and over responsiveness of immune system
- Under responsiveness → uncontrolled infection
- Over responsiveness → hypersensitivity rxn

- Our **genetic diversity** → variations in our ability to mount inflammatory response
- Adaptive immune response to infectious agent → defense
- Adaptive immune response to **non infectious Ag** → **disease**

Three types of non infectious Ags

1. Innocuous (harmless) Ag → hypersensitivity disease
2. Self Ag → autoimmune disease
3. Transplanted Ag → graft rejection

Allergy and hypersensitivity diseases

- 4 types
 1. Type I
 2. Type II
 3. Type III
 4. Type IV

Allergy (Type I hypersensitivity disease)

- Immediate type HSR → rapidly developing reaction
- Mediated by IgE Ab
- Atopic individuals → prone to allergy

- The Ag → Allergen
 - small highly soluble proteins
 - carried by dry particles e.g. pollen grain, house dust etc.
 - most are inhaled
 - most enter transmucosal at low dose
- The immune response → allergic reaction

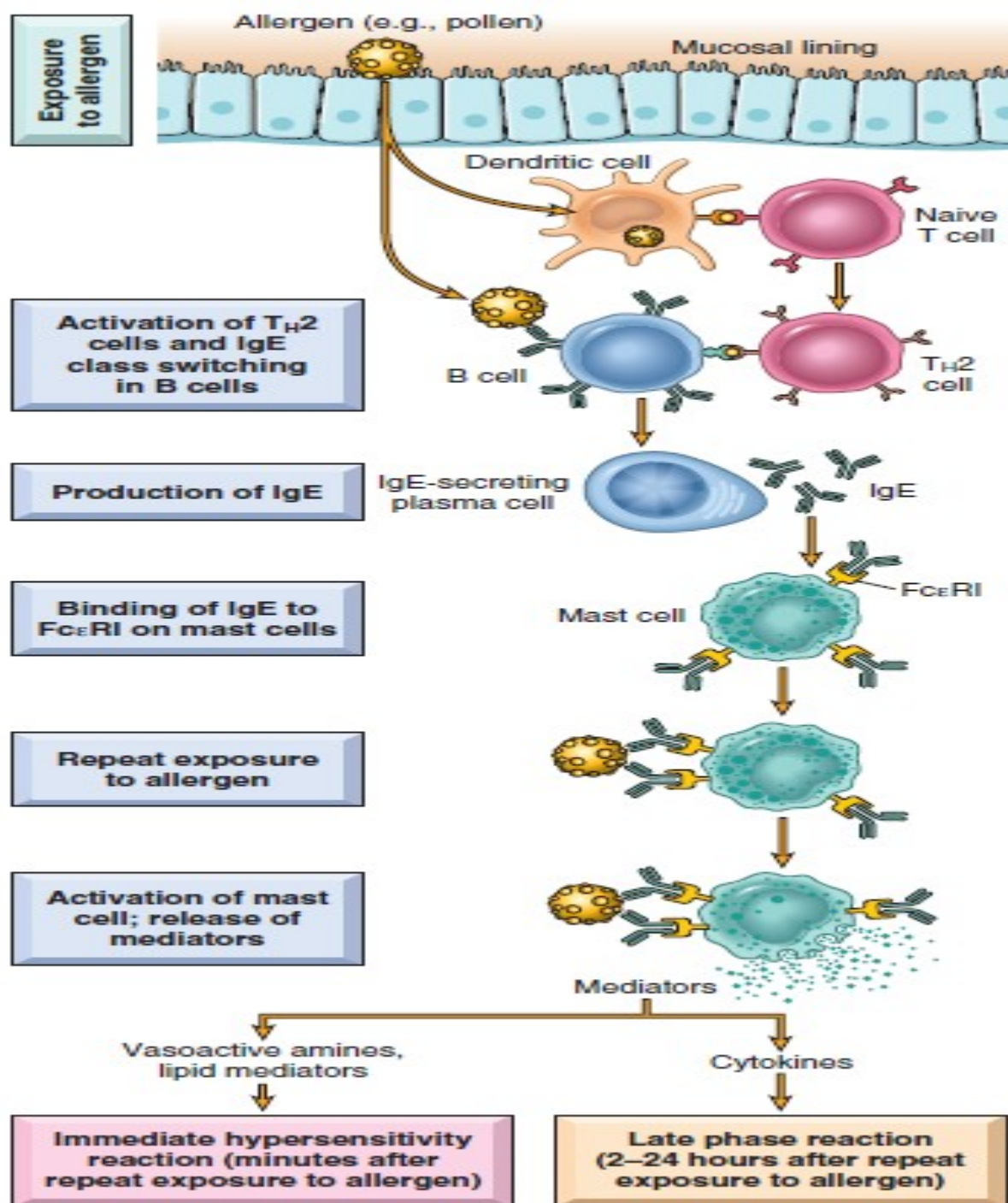
Genetic factors in allergy

- High familial incidence
- Exaggerated tendency to form IgE Ab to allergens (atopy)

- Atopic individuals
 - higher IgE levels in circulation
 - higher level of eosinophils
 - higher affinity IgE receptors on mast cells

Mechanism

- 1st exposure to allergen→
 - *Activation of TH2 cells and production of IgE antibody*
 - *Sensitization of mast cells by IgE antibody.*
- Inhaled Ag→ Contact with mucosa→ Allergen diffuse into mucosa→ Dendritic cells take up allergen and migrate to regional lymph nodes→ Present Ag to lymphocytes (Th2)→ Activate B lymphocytes→ IgE produced→ IgE bound to mast cells



- 2nd exposure→
- Ag+ mast cells armed with IgE→
- Release of chemical mediators
 - degranulation (within seconds)
 - synthesis and release of new mediators

Degranulation

- Release of mediators

1- Vasoactive amines

Histamine

Vasodilation → Hypotension

Smooth muscle contraction, increased mucus secretion

Chemotactic factors- may damage tissue

Influx of inflammatory cells → amplify inflammation

Synthesis and release

2- Lipid mediators

- Prostaglandin D2 (PGD2)- intense bronchospasm and increased mucus secretion.
- The leukotrienes LTC4 and LTD4 are the most potent vasoactive and spasmogenic agents known

3- *Cytokines*- important for the late-phase reaction. These include TNF and chemokines, which recruit and activate leukocytes

Clinical presentation

- Depends on
 1. Amount of IgE present
 2. Route of allergen entry
 3. Dose of allergen

1. Allergen via blood vessels or rapidly absorbed from gut

- mast cells associated with all blood vessels activated
- systemic anaphylaxis (wide spread vasodilatation)
- can be fatal

Anaphylactic shock

- Decreased blood pressure → widespread vessel dilatation and permeability
- Difficulty in breathing → airway constriction
- Suffocation → swelling of epiglottis
- Causes → drugs e.g. penicillin, insect bite (venom), foods e.g. nuts
- Prognosis :rapidly fatal

2. Allergen via inhalation

- most common route

→ **Rhinitis** -mast cells under nasal mucosa
-increased secretion, sneezing

→ **Asthma** -mast cells in lower airway
-bronchial constriction
-increased secretion
-breathing difficulty

3. Oral administration

- Mast cells in GIT
- Vomiting, diarrhea

4. Subcutaneous (skin)

- Skin mast cells
- Large red skin swelling (urticaria or hives)

Type II HSD

- **caused by antibodies directed against target antigens on the surface of cells or other tissue components**
- The antigens may be normal molecules intrinsic to cell membranes or in the extracellular matrix, or they may be adsorbed exogenous antigens (e.g., a drug metabolite).
- Involving **IgG Ab**
- Commonly involves circulating blood cells → **RBC, platelet**

- *Mechanisms of Antibody-Mediated Diseases*
- Antibodies cause disease by targeting cells for phagocytosis,
- activating the complement system, or
- Interfering with normal cellular functions

Opsonization and phagocytosis

- When circulating cells, such as red blood cells or platelets, are coated (opsonized) with autoantibodies, the cells become targets for phagocytosis by neutrophils and macrophages
- Transfusion reaction
- Autoimmune hemolytic anemia

Inflammation

- Antibodies bound to cellular or tissue antigens activate the complement system by the classical pathway
- Complement activation- recruit neutrophils and monocytes, triggering inflammation in tissues.
- Antibody-mediated inflammation is responsible for tissue injury in some forms of glomerulonephritis, vascular rejection in organ grafts, and other disorders.

Antibody-mediated cellular dysfunction.

- antibodies directed against a host protein impair or dysregulate important functions without directly causing cell injury or inflammation.
- In myasthenia gravis,
- In Graves disease,

Type III HSD

- Mediated by **antigen - antibody complexes** formed in blood or at extracellular sites
- Antigen: endogenous or exogenous
- Synonym → **immune complex disease**
- **Deposition of** Ag-Ab aggregates (immune complex) → **inflammation**

- Immune complex formed in all Ab response
- Large complexes → **Cleared** from circulation by macrophages

When do they cause disease?

- Ag excess→
- Small complexes→
- Deposited in blood vessel wall→
- Activate leukocytes→
- Tissue injury

Favored site of immune complex deposition

- Blood vessels
- Glomeruli
- Joints
- Skin
- Heart etc.

Consequences

- Blood vessel → acute **necrotizing vasculitis**
- Glomeruli → **glomerulonephritis**
- Joint → **arthritis**

Type IV HSD

- Synonym → Delayed type reaction
- Mediated by T lymphocytes (TH1, cytotoxic T cells)
- Mechanism similar to response to infectious pathogens e.g. TB
- Reaction to non infectious substance → Hypersensitivity

Autoimmune diseases

- Tissue damage **by adaptive immune response** to self Ag
- Caused by:
 - auto antibodies
 - autoimmune **T-cells**
- * *Self Ag impossible to remove completely*
 - **sustained inflammation**

- Classified into

1. Organ specific:

- Usually endocrine gland
- Hormone receptors or intracellular enzymes

2. Non-organ specific:

- Multiple organs
- Intracellular molecules involved in transcription & translation

- **Organ specific:** Hashimoto's thyroiditis, Graves' disease, pernicious anemia, myasthenia gravis, type I diabetes mellitus, multiple sclerosis, etc
- **Non-organ specific:** Systemic lupus erythematosus (SLE), rheumatoid arthritis, systemic sclerosis, etc
- Some autoimmune diseases can appear in the **infants** of affected mothers as consequence of **transplacental Ab (IgG) transfer**

Autoimmune disease

Autoimmune diseases transferred across the placenta to the fetus and newborn infant		
Disease	Autoantibody	Symptom
Myasthenia gravis	Anti-acetylcholine receptor	Muscle weakness
Graves' disease	Anti-thyroid-stimulating hormone (TSH) receptor	Hyperthyroidism
Thrombocytopenic purpura	Anti-platelet antibodies	Bruising and hemorrhage
Neonatal lupus rash and/or congenital heart block	Anti-Ro antibodies Anti-La antibodies	Photosensitive rash and/or bradycardia
Pemphigus vulgaris	Anti-desmoglein-3	Blistering rash

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Autoimmune disease

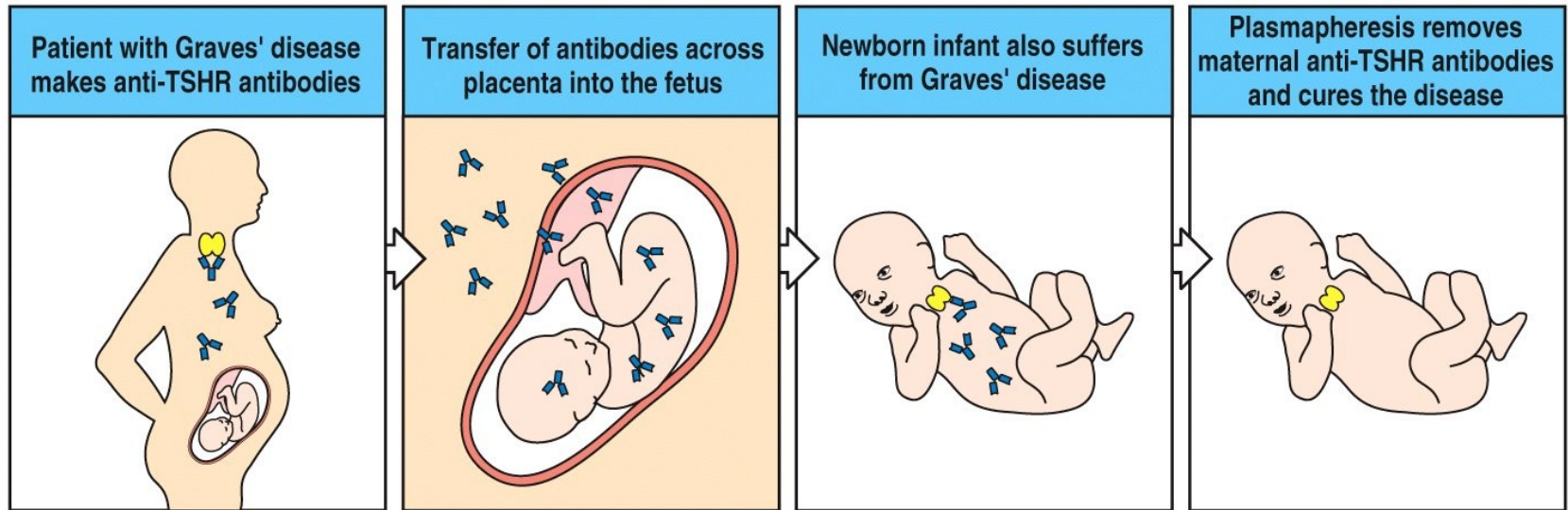


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Epidemiology of Autoimmune disease

- 3%
- 15-65yrs
- More common in women

Mechanism

- All types of HSD except Type I
- Direct antibody mediated effects (type II)
- T cell mediated effects (cellular immune, type IV)
- Immune complex mediated effects (type III)

Self tolerance

- **Central Tolerance** - this occurs during lymphocyte development.
- **Peripheral Tolerance** - occurs after lymphocytes leave the primary organs.

Predisposing (aetiologic) factors

- ❑ Interaction between genetic & environmental factors are important

1. Genetic

- Studies on familial clustering & monozygotic twins
- Major histocompatibility complex (MCH) genotype: HLA associations e.g., HLA B27

2. Environmental - infections (viral & bacterial), drugs, UV

- Mechanisms:

A. Co-stimulatory receptors

- Infectious agents => expression of co-stimulatory receptors on Ag presenting cells (APC). Now these APC can present self Ag to auto reactive T cells

B. Molecular mimicry

- Abs or T cells generated in immune response to infectious agent cross-reacts with self Ag

Examples:

□ Rheumatic fever:

- Group A streptococci pharyngitis => Ab against streptococci => Ab cross react with kidney, joint & heart Ags => Type II inflammatory reaction => Disease

□ Myasthenia Gravis

- Ab to glycoprotein D of **herpes simplex virus** => cross react with acetylcholine receptors => myasthenia gravis

Predisposing (aetiologic) factors

3. **Hormonal factors** - common in WOMEN, peak incidence in child-bearing age

Mechanism of Autoimmune disease

- Failure of self tolerance=> Autoantibodies (Type II & III Rxn) & autoimmune T cells (type IV Rxn) => tissue damage => autoimmune disease

Type II Rxn

IgG or IgM against self Ag on **surface**
of cell or **extra cellular matrix**

Examples

❑ Autoimmune hemolytic anemia

- Self Ag on RBCs + Ab => RBCs destruction

❑ Autoimmune thrombocytopenia

- Self Ag on platelets + Ab => platelets destruction

❑ Good pasture's syndrome

- Self Ag on basement membrane (BM) collagen + Ab
- BM of **glomeruli** => vasculitis & renal failure
- BM of **alveoli** => pulmonary hemorrhage

Examples

B. Auto antibodies to cell surface receptor

□ Stimulate the receptor => Grave's Disease

- Autoantibodies to thyroid stimulating hormone (TSH) receptor on thyroid cells => excess T3 & T4
=> Hyperthyroidism

□ Block the stimulation => Myasthenia gravis

- Autoantibodies to acetylcholine receptor at neuromuscular junction => progressive weakness

Mechanism of Autoimmune disease ctd

2. Type III immune complex (IC) disease

- Ag excess => IC disease

E.g. Systemic lupus erythematosus (SLE)

=> Autoantibody to self DNA

=> Large quantities of Ag (DNA) available

=> Small sized IC formed

=> IC deposition in small blood vessels, glomeruli & joint => activation of complement => tissue damage

Mechanism of Autoimmune disease

3. Type IV (T cell mediated disease)

- Auto-reactive T cells => inflammation

□ Rheumatoid arthritis

- T-cells specific for joint Ag => arthritis
- Auto-Ab (IgM anti-IgG autoantibody called rheumatoid factor)

□ Type I DM

- Auto Ag on beta cell of pancreatic islet of Langerhans. Auto reactive CD8 T-cells destroying the beta cells. CD4 T cells has also role.

Clinical e.g.

- Insulin dependent diabetes mellitus
 - Auto Ag= protein expressed on insulin producing B cells of pancreatic islet of Langerhan
 - Destroyed by auto reactive CD8 T cells

